

Assignment #3 – AI System Implementation Analysis

I. How AI behaviors influence player strategy and decision-making

AI behavior plays a crucial role in shaping how players approach challenges within a game. Through state transitions, players are able to analyze enemy behavior and make strategic decisions, such as moving out of detection range to find a safe area. The use of animation changes to reflect different AI states provides clear visual feedback to the player, helping them understand the enemy's current condition.

Additionally, incorporating unique sound cues for each state transition can serve as auditory hints or "warnings," giving players time to react accordingly. This kind of feedback system not only enhances immersion but also allows players to anticipate behavior changes, encouraging more thoughtful and tactical gameplay.

II. How player actions dynamically alter AI states and responses

In my implementation, I used proximity-based detection, where the AI monitors a radius around itself to identify when a player is nearby. Once the player is detected, the AI transitions into a new state, which is visually communicated through an animation change. This provides players with a brief response window to adapt their behavior.

While I attempted to implement an audio cue system to complement the visual changes, I encountered technical challenges that prevented me from completing it. However, adding audio feedback is a valuable layer of interaction that gives players an additional way to interpret AI behavior and respond effectively. These types of dynamic interactions help create a more immersive and responsive environment, making the gameplay experience feel smoother and more human-like.

III. Challenges faced during implementation and their solutions

Programming has never been my strongest skill. Although I genuinely enjoy creating these types of mini-games, I often feel frustrated by the limitations of my technical knowledge. When I want to add a new mechanic, it usually requires extensive research and time spent watching tutorials or asking for help. This can slow down the development process and create a gap between the idea and its execution.

One specific example was my failed attempt to integrate audio transitions alongside AI state changes. I found the concept both fun and meaningful, but due to time constraints and lack of experience, I wasn't able to finish it. Despite these difficulties, I truly enjoyed watching my idea take form through code. Seeing AI behaviors come to life through small systems reminds me of why I love game design, even if the process is challenging. I see these experiences not as setbacks, but as learning opportunities that push me to grow and improve with every project.